Datathon 2026 - Urban Flow Analytics Data Challenge

Challenge Overview

Urban transportation networks generate millions of trip records containing information about journeys, fares, locations, timestamps, passengers, and payment methods.

However, raw operational data is rarely perfect. It may contain invalid records, unusual values, inconsistencies, and patterns that are difficult to identify without careful analysis.

As a data science team, your task is to analyze the Urban Flow Analytics Data, assess its quality, uncover meaningful patterns, and build predictive solutions that can help improve taxi operations.

Dataset Description

Two datasets are provided for this data challenge in **CSV format**:

1. **Urban Flow Analytics Taxi Dataset**
   This dataset contains trip level taxi records, including journey timestamps, pickup and drop off locations, trip distance, passenger count, fare details and other relevant operational attributes. Participants should use this dataset for data quality assessment, exploratory analysis, predictive modelling, demand forecasting, and spatial-temporal analysis.
2. **Urban Flow Analytics Zone Dataset**
   This dataset contains reference information about the geographical taxi zones. It can be joined with the taxi dataset using the relevant pickup and drop-off zone identifiers to determine the corresponding location details. This dataset supports location-based analysis, hotspot identification, origin-destination flow analysis, and visualisation of travel patterns.

A separate **data dictionary** is attached for each dataset. The data dictionaries provide column names, data types, and descriptions of the available fields. Participants are advised to review them before beginning data preparation and analysis.

Problem

1. Exploratory / Data Quality

Data cleaning challenge: Identify and handle anomalies - negative fares, trip distances of 0 with nonzero fares, passenger counts of 0, trips with drop-off before pickup, unrealistic speeds (distance/time).

For every anomaly type, clearly justify whether you chose to drop, impute, or filter the rows, and quantify what percentage of total rows were impacted. Include the justification in your notebooks.

2. Prediction & Supervised Learning

2.1 Fare Amount Prediction:

To build passenger trust, the city is launching a "No-Surprises Upfront Pricing" feature that locks in a ride's price before you get into your taxi.

Develop a machine learning pipeline to predict base_fare prior to the start of a trip to power this real-time pricing engine.

2.2 The On-Time Arrival Estimator:

City traffic changes quickly, making simple distance formulas inaccurate. Passengers need to know exactly when they'll reach their destination before setting off.

Build a model to accurately predict total trip time down to the minute.

3. Spatial-Temporal & Demand Analytics

3.1 The Fleet Dispatcher

Taxi drivers waste time and fuel going around empty streets when they don't know where passengers are waiting. Fleet dispatchers need to predict passenger demand ahead of time so drivers can position themselves in the right neighborhoods.

By building time-series models for top taxi zones, predict pickup volumes for the next 24 to 72 hours to keep taxis moving and wait times low.

3.2 Hotspot & Origin-Destination Flow Clustering

City planners need to see where people travel most so they can manage. Looking at millions of single taxi rides makes it hard to see the big picture.

By clustering pickup and drop-off patterns, map out major travel hotspots to show how movement changes from morning work hours to late at night.

4. Architecture & Comprehensive Technical Report

All teams must submit a detailed technical report covering:

- Data preprocessing and feature engineering (limit to 1 page)
- Model development methodology.
- Evaluation metrics and results.
- High-level key findings, business implications and conclusions.
- Solution Architecture Diagram illustrating the complete ML pipeline from data input to prediction.

  (Total page limit: 20 pages)

Bonus Points

Track 5: The AI Mobility Assistant
Most city officials and fleet managers are unfamiliar with how to write complex SQL or Python code, so they need fast answers from millions of taxi records. Instead of waiting for data analysts to write custom reports that can take days, they prefer to ask questions in plain English.

Build an intelligent question answering agent that can safely translate natural questions to code, retrieve immediate answers and handle ambiguous or malformed questions gracefully.

Track 6: Turning Taxi Data into Business Decisions

A taxi ride company generates a large amount of data from its daily operations — including trips, fares, locations, timings, payment methods, tips, and trip durations. However, raw data alone does not tell the business what problems exist or what actions should be taken.

Challenge

1. Identify a Business Problem or Opportunity
   Explore the available taxi data and formulate a specific problem, question, or hypothesis that could have a meaningful impact on the company's business.

2. Analyze the Data
   Use appropriate analytical techniques to investigate the problem, identify patterns and trends, and uncover insights that support or challenge your identified business problem.

3. Build an Interactive Business Dashboard
   Develop an interactive dashboard designed specifically for the taxi company's management. The dashboard should help the taxi company monitor the identified problem, explore the underlying factors, and make informed decisions. The primary audience for the dashboard is the taxi company's management and business decision-makers, such as operations managers, business analysts, and senior management.

4. Tell a Data-Driven Story
   Your dashboard should not simply display charts and numbers. It should tell a clear story:

What is the problem? → What does the data tell us? → Why is it happening? → What should the business do about it?

5. Recommend Actions
   Based on your analysis, provide practical and data-backed recommendations that the taxi company could implement to improve its operations, revenue, customer experience, efficiency, or overall business performance.

<u>Possible Areas to Explore</u>

Teams are free to identify their own business problem. Some possible areas include:

- Revenue and profitability optimization
- Identifying high- and low-performing locations
- Understanding demand patterns and peak periods

- Understanding factors affecting fares and trip revenue
- Identifying opportunities to increase tips or other revenue
- Understanding payment-method trends

These are examples only. Participants are encouraged to discover a problem that they believe is important and demonstrate why it matters to the business.

The final solution should demonstrate that data can be transformed into insights, insights into decisions, and decisions into business value.

Evaluation

The evaluation will focus on the effectiveness of data preprocessing, feature engineering, and the predictive performance of the machine learning model using the Urban Flow Analytics datasets.

Model Evaluation Aspects

Teams are encouraged to evaluate their models using the following aspects:

- Data Preprocessing & Feature Engineering: Assess how effectively missing values, outliers, categorical variables, datetime features, and location-based information were processed to improve model performance.
- Prediction Accuracy: How accurately does the model predict the target variable (e.g: trip fare, trip duration)
- Regression Performance: Evaluate the robustness of the model using appropriate regression metrics such as RMSE, MAE, and R^2 Score.

Rationale for Selection of Model Evaluation Metric/s

As part of your submission, include a brief explanation of the evaluation metrics you selected.

Address the following:

- Reasoning for metric selection: Explain why the chosen metrics are appropriate for your predictive modeling approach.
- Relevance to the challenge: Describe how the metrics measure prediction accuracy and overall model effectiveness.

- Interpretability: Explain how the selected metrics help compare different models during experimentation.

Grading of Model Evaluation Method

Your evaluation methodology and justification will be assessed based on:

- The effectiveness of the preprocessing and feature engineering strategy in improving predictive performance.
- Demonstration of an appropriate approach to hyperparameter tuning and model optimization.
- The appropriateness and justification of the selected evaluation metrics.
- The quality of model validation and comparison.

Submission

Each team must submit the following:

1. Detailed Technical Report
2. Source Code including all notebooks (.ipynb) and supporting scripts used to build the solution. Include Exploratory Data Analysis with a brief explanation in your notebook.
   The final notebook must be named **TeamName_FinalNotebook.ipynb**.
3. Trained Model in .pkl format.
4. Dataset with clearly separated training, validation, and testing splits.
5. Demo Video (3–5 minutes, unlisted YouTube) explaining the solution, architecture, methodology, and results.

Submission format:

Compress all deliverables into a single ZIP file named:
TeamName_CodefestDatathon2026.zip

Rules and Regulations

Teams need to complete sections 1–4, which will contribute to the shared leaderboard.

The remaining 5–6 sections are open-ended and intended for creativity, innovation, and bonus points.

Each team may submit only one final solution. The latest submission before the deadline will be evaluated.

Selected teams may be invited to a 5-minute discussion/interview with the judges to explain their solution architecture, methodology, and implementation. This discussion will be used to understand the submitted solution.

Any plagiarism, unauthorized collaboration, or rule violations will result in disqualification.

The judges' decisions are final and binding.

Terms & Conditions

Dataset Usage: The provided datasets are to be used exclusively for SLIIT Codefest Datathon 2026.

Confidentiality: Teams must not publish, redistribute, or publicly disclose the competition datasets or their derivatives without authorization from the organizers.

Original Work: All submitted reports, code, models, and documentation must represent the team's own work.

Reproducibility: The submitted code and notebooks should be executable and capable of reproducing the reported results.

Violation of Terms: Any breach of these terms may result in disqualification and any further action deemed necessary by the organizers.